

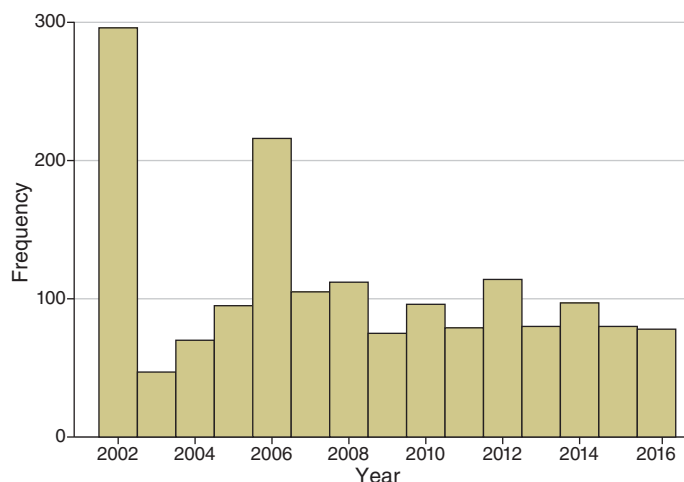


FIGURE 1. NUMBER OF MUNICIPALITIES RECEIVING A CAPS (OF ALL TYPES) BY YEAR

Notes: This graph plots the number of municipalities receiving a CAPS for the first time from 2002 to 2016. A number of CAPS were created before the period and were accredited in 2002, explaining the spike in 2002. For the remaining CAPS, the accreditation coincides with opening. These data show the first date of accreditation, or date of CAPS' opening for the vast majority. We ignore closures of CAPS since only 13 were closed during this period.

30,000 and 100,000 BRL per month for maintenance. Figure 1 shows the number of municipalities that received a CAPS each year from 2002 to 2016. Since most municipalities in Brazil are relatively small, it is unsurprising that the vast majority of CAPS are CAPS I. In Figure A1, we show how CAPS spread across the country over time. The rollout does not seem to have followed any clear geographical pattern.

Despite the formal guidelines and requirements, local administrations face many challenges in effectively implementing CAPS. Even with financial support from the federal government, many municipalities discover that more money is needed to maintain their CAPS in an ideal state. A lack of resources is particularly notable in terms of professionals, especially physicians. Doctors are typically hired part-time, and they work in the CAPS just a few times per week. On the one hand, this facilitates the hiring process, since even doctors already working in the private sector can fit the CAPS work into their schedules. On the other hand, limited availability of physicians may prevent patients from receiving proper treatment.

It is also important to distinguish the most common types of CAPS in Brazil from the community mental health centers (CMHC) in the United States. Since CAPS are usually much less well equipped, are not open 24 hours, have no beds, and can lack psychiatrists most of the time, they tend to be ill-

suited to treat patients who need a higher level of care. CAPS III is a type of center that is more similar to the CMHCs, but only a handful exist in Brazil.

Another key piece of information is related to the type of patients treated at CAPS. Online Appendix Table A1 compares patients treated in CAPS to those who report having been diagnosed with a mental illness in a national survey conducted in 2019 (Instituto Brasileiro de Geografia e Estatística 2019). Although the survey only asks about a limited number of conditions, comparing the responses to the data

from CAPS uncovers a clear pattern. Most people diagnosed with mental illnesses in Brazil suffer from depression (76 percent), while only a tiny fraction have schizophrenia (6 percent), a much more severe condition. In contrast, only 12 percent of people treated in CAPS suffer from depression, while schizophrenia is the most common illness treated in the centers (28 percent of the patients).

On the one hand, this profile aligns with policy guidelines that specify that CAPS should provide treatment to individuals with medium- to high-severity mental illnesses. On the other hand, the profile of the typical patient may not be consistent with the resources available in the centers. Because of this mismatch, some specialists and families of severely ill patients in Brazil favor a model that includes psychiatric hospitals (e.g., Filho 2021; G1-BA 2017; Pinheiro 2019; Teixeira and Coelho 2019).

Along similar lines, we must understand how CAPS are related to the other aspect of the psychiatric reform: a reduced number of psychiatric beds. CAPS are explicitly intended to be substitutes for psychiatric hospitals. Thus, it is not unreasonable to assume that acquiring CAPS and closing psychiatric beds are intrinsically and strongly related processes. However, this is not what happens in Brazil. Even before the psychiatric reform law, the idea that mental health care should not be primarily inpatient was already gaining traction. Since the 1980s, Brazil has experienced a significant reduction in the total number of psychiatric beds, under the assumption that primary care is more appropriate than inpatient care. However, policymakers have attempted to ensure that this transition be gradual, and that a sudden loss of psychiatric beds would not disrupt local mental health care systems. They advocate that municipalities should be given guidelines to reduce inpatient psychiatric capacity efficiently, regardless of the presence or absence of CAPS. We investigate the relationship between CAPS and psychiatric beds and report our findings in Section IV.

B. Theory of Change

Here, we begin to investigate how CAPS impact outcomes of interest. We discuss the theory of change and defer to the next section our assessment of measurement and data limitations. Figure 2 summarizes our understanding of the impact of CAPS on our outcomes of interest.

When a CAPS is established, it becomes the primary source of care for individuals with mental illness. The goal is to replace hospitals. The center also increases availability of outpatient services. As a result, we may expect changes in the allocation of mental health care resources in this new model of care.

These process changes should shift the market toward a new equilibrium, with different levels of community- and hospital-based mental health care capacity. In particular, the policy should influence (i) the decision to seek treatment or not, and (ii) conditional on seeking treatment, which type of treatment patients receive. After the reform, we expect that some people would switch from their original choice (seek outpatient care at public clinics, seek inpatient care, or not seek treatment at all) toward outpatient care at CAPS, and that other health units would increase referrals to CAPS. Given the goals of the reform, we focus on the substitution of inpatient with outpatient care.

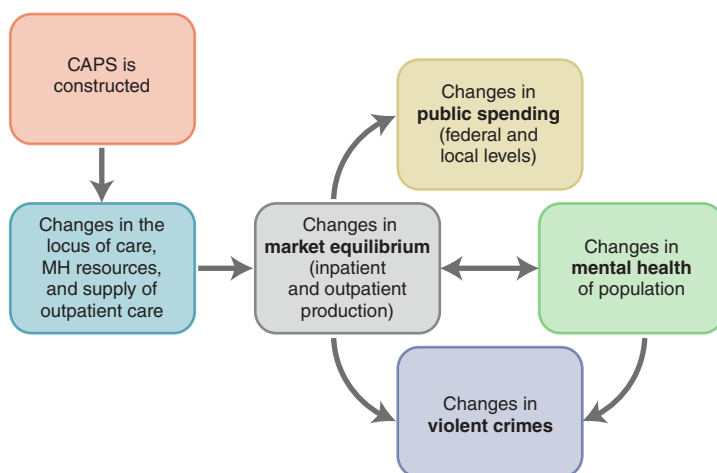


FIGURE 2. THEORY OF CHANGE

Notes: This diagram summarizes the changes we expect to happen following the construction of a CAPS in a given municipality. “MH” stands for mental health.

A new equilibrium may affect public spending related to mental health. This is particularly important, since one of the rationales behind the reform is its cost-saving potential, compared to inpatient mental health care delivered through hospitals. It may also change mental health outcomes. The consequences could be positive or negative for patients, depending on how well outpatient care suits each case, compared to inpatient care. If the effect on mental health is large enough, it could affect mortality rates, especially those related to serious illnesses (Cavanagh et al. 2003; Olfson et al. 2015).⁶

Individuals with more serious mental illnesses are prone to violent behavior and victimization (Brekke et al. 2001a; Teplin et al. 2005a; Fazel and Grann 2006; Rueve and Welton 2008). Inpatient treatment may help lower violent crime through incapacitation. The overall effects of the reform depend on its impact on people’s mental health, and whether CAPS meet the treatment needs of individuals with severe mental illnesses. Analyzing the rate of violent crimes is particularly pertinent given the accounts of violence related to CAPS patients reported by the media (e.g., Sousa 2012; G1-Sorocaba/Jundiaí 2016; Vilela 2018; Vaccari 2019).

II. Data

Most of the data we use come from the Brazilian Ministry of Health. We combine records of deaths, hospitalizations, outpatient procedures, and health facilities from all over the country. Since the policy is implemented at the geographical level of municipalities, we use them as our unit of analysis.⁷

⁶Due to data limitations, we are unable to assess how finer, less extreme mental health measures are affected.

⁷Since some cities were founded during the period of study, we aggregate them into minimum comparable areas using data from the Institute of Applied Economic Research (IPEA) (Instituto de Pesquisas Econômicas e Aplicadas

The Brazilian Ministry of Health provides the data regarding the opening of each CAPS (Ministério da Saúde do Brasil 2019).⁸ The data cover the period between 2002 and 2019. Included are the opening date and type of every CAPS created after the psychiatric reform law was sanctioned. For the few CAPS that predate this period, the data indicate 2002 as the year of opening. Hence, we only exploit variation from 2003 onwards. Most municipalities with a center have a CAPS I (80 percent) and adopted only one center during this period.⁹

To assess the effects of CAPS on the supply of outpatient services, we use administrative data from the National Registry of Health Establishments (*Cadastro Nacional de Estabelecimentos de Saúde*—CNES) (DATASUS 1979–2024a). This registry includes detailed data on all public and private health facilities in Brazil. It has information about health professionals linked to health care facilities, their area of practice, and levels of specialization. For each municipality, we select the number of mental health providers that usually constitute community mental health teams: psychiatrists, psychologists, occupational therapists, and social workers. These providers represent, on average, 87 percent of all the professionals working at the psychosocial care centers. The data also provide information on the number of hospital beds in each municipality, and we select those used for psychiatric patients according to the registries.

To study the effects of CAPS on the equilibrium of the mental health care market, we use two datasets: one for outpatient care and the other for inpatient care. The first is the National System of Information on Ambulatory Care (*Sistema de Informações Ambulatoriais de Saúde*—SIA) (DATASUS 1979–2024d). SIA provides administrative information on all visits funded by the public sector in which medical care is provided on an outpatient basis. Visits may take place in any facility that provides primary health services. Severe compatibility issues limit the usefulness of this dataset. Its data are at the procedure level, and many procedure codes change over time, making comparisons impossible. With few exceptions, therefore, we avoid evaluating specific outpatient procedures. From 2008 onward, we can identify the type of health professional providing outpatient care. We then select the overall number of outpatient services provided by each type of mental health provider we are evaluating. SIA also contains information on “outpatient pharmaceutical assistance” regarding medications provided to patients. We focus on antipsychotic medications distributed in each municipality.

Regarding inpatient care, the Hospital Information System of the Unified Health System (*Sistema de Informações Hospitalares*—SIH) provides information about hospital admissions that use beds in the public health care sector (DATASUS 1979–2024e). This dataset includes information on admissions by patient’s municipality of residence and by cause of admission, coded using the ICD-10. We rely on the ICD classification to identify causes of hospitalization related to mental illnesses

1995–2004). For the sake of clarity, we refer to minimum comparable areas as municipalities.

⁸These data were requested directly to the Ministry of Health through the Information Access Law. These data are not formally published anywhere as far as we know, but are public nonetheless and are provided with the rest of our data.

⁹For more details on the last two points, see online Appendix Figures A2 and A3, respectively.